Free Daily Utility Tools

- **Office tools:** Apache OpenOffice (*www.openoffice.org*)
- **Zipping software:** WinZip (*www.winzip.com*)
- **Browser:** Mozilla Firefox (*www.mozilla.org*)
- **PDF Reader:** Adobe Acrobat Reader DC (*get.adobe.com*)
- **Media player:** PotPlayer (*potplayer.daum.net*)

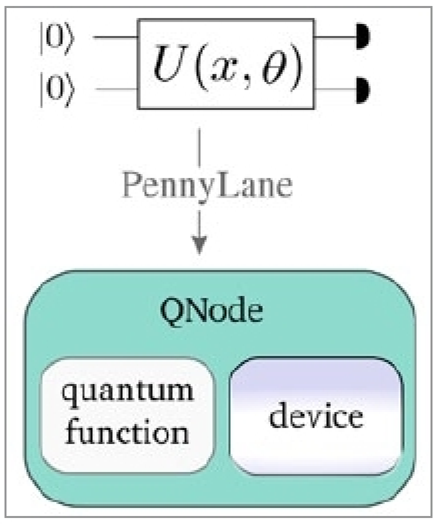
Software Tools For QUANTUM COMPUTING

Ayushee Sharma

PennyLane

Visit: *https://www.xanadu.ai/software*
Full version: Free

PennyLane is a free, open-source, device-independent cross-platform Python library from Xanadu that specialises in quantum machine learning (ML) and optimisation of hybrid quantum-classical computations. It consists of built-in automatic differentiation of variational quantum circuits (like Xanadu's Strawberry Fields and Google's Cirq). NumPy is the standard interface with other options for interfacing like PyTorch and Tensorflow available too, thereby providing support for both hybrid quantum and classical models. It works with Python versions 3.5 and above.



ScaffCC

Visit: *https://github.com/epiqc/ScaffCC*
Full version: Free

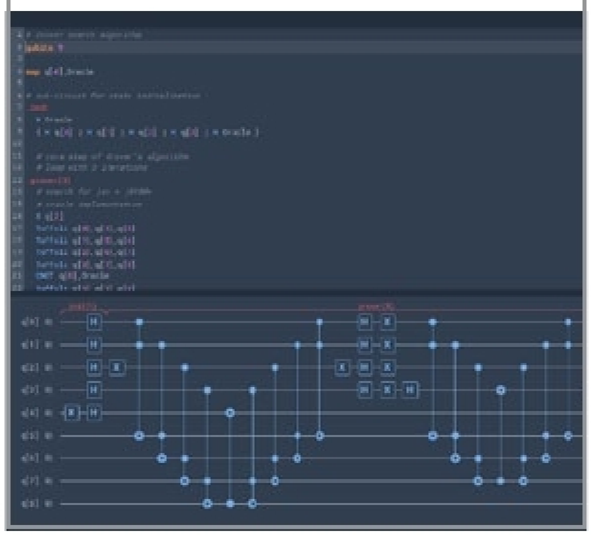
ScaffCC is an open-source compilation, analysis and optimisation framework for the high-level programming language Scaffold, which is used for quantum algorithms. The software comes with built-in quantum application for ground state estimation of a molecule, secure hash algorithm (SHA 1), Shor's factoring algorithm, Quantum Fourier Transform (QFT), among others. It allows researchers to compile Scaffold algorithms to flat quantum assembly format, and transform that output into an acceptable format for the QX quantum computer simulator. It currently offers support for Ubuntu, Red Hat, and OS X operating systems.

```
#define s_ 3000     // iteration count

module Oracle (qbit a[1], qbit b[1], int j) {
  double theta = (-1)*pow(2.0, j)/100;
  X(a[0]);
  Rz(b[0], theta);
}

module main () {
  qbit a[1], b[1];
  int i, j;
  for (i=1; i<=s_; i++) {
    for (j=0; j<=3; j++) {
      Oracle(a, b, j);
    }
  }
}
```

Quantum Inspire

Visit: *www.quantum-inspire.com*
Full version: Paid

Quantum Inspire (QI) by QuTech is a quantum computing platform that allows users to perform quantum computations efficiently. Users can program quantum algorithms in quantum assembly language, execute them, examine the results using a built-in data viewer and download the output. Features like automatic bug identification and auto complete in QI editor are available to aid non-experts. The editor also displays the list of available quantum bit (qubit) operations and circuit visualiser, which automatically gets rendered from the code in real time.



Quirk

Visit: *https://algassert.com/quirk;* **Full version:** Free

Quirk is a quick and interactive open source drag-and-drop simulator for small quantum circuits. It comes with examples (circuits) to help understand quantum algorithms, experiments and the overall functionality of circuits. Several gates (that may be split sometimes) are included for performing arithmetic operations. Simple actions like saving circuits, gate dragging, duplication and deletion, qubit addition, gate alternation and more are available for easy editing. Display objects (chance, amplitude, density matrix, bloch sphere) are useful to know about the quantum state at a particular time and place.